

UNIVERSITY OF RUHUNA
BACHELOR OF INFORMATION AND COMMUNICATION TECHNOLOGY DEGREE
LEVEL II (SEMESTER I)

SOFTWARE REQUIREMENTS SPECIFICATION

Faculty of Technology Management System

ICT2132 – Object Oriented Programming Practicum

Mini Project

GROUP NO -20

TG/2024/2114 - K.P.S.B PATHIRANA

TG/2114/2087 - V.D.M PERERA

TG/2024/2141- R.G.M.S RATHNAYAKA

TG/2024/2112- G.M.S SARATHCHANDRA

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of the proposed Faculty of Technology Management System. The system will be developed using Java and MySQL and will manage user profiles, course details, undergraduate marks, attendance, notices, timetables and medical records.

1.2 Scope

The system will provide facilities for managing user profiles, courses, course materials, undergraduate marks, attendance, medical records, notices, timetables, eligibility, grades, SGPA and CGPA.

1.3 Intended Users

The intended users are Administrator, Lecturer, Technical Officer and Undergraduate Student.

1.4 Product Overview

The proposed system will be a Java-based GUI application connected to a MySQL database. It will provide role-based access so that each user can access functions according to their responsibilities.

2. Overall Description

2.1 Product Perspective

The proposed system is an academic management system for the Faculty of Technology. It will provide a centralized database and reduce manual handling of academic records.

2.2 Main Components

The main components are User Management, Course Management, Attendance Management, Marks Management, Medical Management, Notice Management, Timetable Management, and Grade/GPA Management.

2.3 User Classes and Characteristics

Administrator

- Create and maintain all User Profiles.
- Create and maintain Courses.
- Create and maintain Notices.
- Create and maintain Timetables.

Lecturer

- Update their profile except username and password.
- Modify and add materials to courses.
- Upload marks for all kinds of exams.
- View undergraduate details and eligibility.
- View undergraduate marks, grades and GPA.
- View attendance and medical records.
- View notices.

Technical Officer

- Update their profile except username and password.
- Add and maintain undergraduate attendance details.
- Add and maintain medical details.
- View notices.
- View timetables of their department.

Undergraduate

- Update only contact details and profile picture.
- View attendance and medical details.
- View course details.
- View grades and GPA.
- View timetables and notices.

3. Functional Requirements

Requirement	Description
Login	The system shall allow users to log in using a username and password.
Authentication	The system shall verify login credentials before granting access.
Role-Based Access	The system shall provide functions according to the user's role.
User Profile Management	The Administrator shall be able to create and maintain user profiles.
Profile Update	Lecturers and Technical Officers shall be able to update their profiles except username and password.
Undergraduate Profile	Undergraduates shall only be able to update contact details and profile picture.
Course Management	The Administrator shall be able to add, update and maintain course information.
Course Materials	Lecturers shall be able to add and modify materials to courses.
Attendance	The Technical Officer shall be able to add and maintain undergraduate attendance.
Attendance Summary	The system shall provide individual and whole-batch attendance summaries.
Medical Records	The Technical Officer shall be able to add and maintain medical details and approval status.
Marks	Lecturers shall be able to upload marks for all applicable examinations.
CA Eligibility	The system shall identify eligibility based on the CA requirement of at least 40%.
Combined Eligibility	The system shall check eligibility using attendance and CA marks.
Grades	The system shall calculate and display grades according to UGC Commission Circular No. 12-2024.
GPA	The system shall calculate and display SGPA and CGPA.
Notices	The Administrator shall be able to create and maintain notices; other users shall be able to view them.
Timetables	The Administrator shall be able to create and maintain timetables; relevant users shall be able to view them.

4. Attendance Management Requirements

The system shall support 15 theory sessions and 15 practical sessions for course modules. It shall support attendance scenarios required by the project specification.

- More than 80% attendance.
- Exactly 80% attendance.
- Less than 80% attendance without medicals.
- More than 80% attendance with medicals.
- Less than 80% attendance with medicals.
- View attendance for theory only.
- View attendance for practical only.
- View combined theory and practical attendance.
- View attendance for an individual student and the whole batch.

5. Marks and Eligibility Requirements

- All marks shall be recorded out of 100.
- The system shall maintain CA marks and final examination marks.
- The system shall provide CA mark summaries for individuals and the whole batch.
- CA must be at least 40% for the CA eligibility requirement.
- The system shall provide individual and whole-batch eligibility checking.
- The system shall provide final marks for individuals and the whole batch.

6. Grade, SGPA and CGPA Requirements

- The system shall grade students according to UGC Commission Circular No. 12-2024.
- The system shall display marks and grades for each subject.
- The system shall display individual and whole-batch grades.
- The system shall calculate SGPA and CGPA.
- The system shall show subjects, grades and GPA information for each undergraduate.

7. Non-Functional Requirements

Category	Requirement
Performance	The system should respond within a reasonable time during normal use.
Security	Only authenticated and authorized users shall access protected functions.
Reliability	Academic records shall be stored accurately and consistently.
Usability	The GUI should be simple, clear and easy to use.
Maintainability	The system should follow OOP principles to support future changes.
Scalability	The system should support additional users, courses, marks and attendance records.

8. Database Requirements

The system shall use MySQL as the database management system.

- Users
- Students
- Lecturers
- Technical Officers
- Courses
- Course Materials
- Attendance
- Medical Records
- Marks
- Grades
- GPA
- Notices
- Timetables

9. GUI Requirements

9.1 Login Screen

- Username
- Password
- Login button

9.2 Admin Dashboard

- User Management
- Course Management
- Notice Management
- Timetable Management

9.3 Lecturer Dashboard

- Profile
- Courses
- Course Materials
- Marks
- Eligibility
- Students
- Attendance
- Medical Records
- Grades and GPA
- Notices

9.4 Technical Officer Dashboard

- Profile
- Attendance
- Medical Records
- Notices
- Timetable

9.5 Undergraduate Dashboard

- Profile
- Courses
- Attendance
- Medical Records
- Grades
- GPA
- Timetable
- Notices

10. Object-Oriented Requirements

The system shall demonstrate the following Object-Oriented Programming concepts:

- Classes and Objects
- Inheritance
- Abstraction
- Polymorphism
- Encapsulation
- Error and Exception Handling
- Database Handling
- GUI

Possible core classes include User, Admin, Lecturer, TechnicalOfficer, Undergraduate, Course, Attendance, Marks, Medical, Notice and Timetable.

11. Security Requirements

- Users shall log in using valid credentials.
- Users shall only access functions assigned to their roles.
- Students shall not be allowed to modify marks or attendance.
- Input data shall be validated before storing it in the database.
- Database access shall be controlled through appropriate authentication and authorization.

12. Initial Data Requirements

The system shall contain minimum sample data required for testing:

- 01 Administrator
- At least 05 Lecturers
- At least 04 Technical Officers
- At least 20 Undergraduates, including repeat and batch-missed students.

13. Use Case Summary

Actor	Main Functions
Administrator	Manage users, courses, notices and timetables.
Lecturer	Manage course materials and marks; view students, attendance, medicals, eligibility, grades and GPA.
Technical Officer	Manage attendance and medical records; view notices and department timetables.
Undergraduate	View courses, attendance, medicals, grades, GPA, timetable and notices.

14. Major Use Cases

Use Case	Actor	Description
Login	All Users	User logs into the system using valid credentials.
Manage User Profiles	Administrator	Creates and maintains user profiles.
Manage Courses	Administrator	Creates and maintains course information.
Manage Course Materials	Lecturer	Adds and modifies course materials.
Manage Attendance	Technical Officer	Records and maintains undergraduate attendance.
Manage Medical Records	Technical Officer	Records and maintains medical information.

Upload Marks	Lecturer	Uploads examination marks.
Check Eligibility	Lecturer/System	Checks eligibility using attendance and CA marks.
View Grades	Lecturer/Undergraduate	Views grades according to permissions.
Calculate GPA	System	Calculates SGPA and CGPA.
Manage Notices	Administrator	Creates and maintains notices.
Manage Timetable	Administrator	Creates and maintains timetables.

15. Assumptions

- Users have valid accounts created by the Administrator.
- Academic course information is provided by the Faculty.
- Lecturers enter accurate marks.
- Technical Officers enter accurate attendance and medical information.
- Students can access only information permitted for them.
- The MySQL database server is available when the application is used.

16. Constraints

- The application shall be developed using Java.
- MySQL shall be used for database management.
- The system shall follow Object-Oriented Programming principles.
- User access shall be role-based.
- Marks shall be recorded out of 100.
- Attendance shall support theory and practical sessions.
- Grades shall follow UGC Commission Circular No. 12-2024.

17. Conclusion

The proposed Faculty of Technology Management System will provide a centralized solution for managing undergraduate academic information. It will manage user profiles, courses, course materials, attendance, medical records, marks, eligibility, grades, GPA, notices and timetables. By using Java, MySQL and Object-Oriented Programming principles, the system will provide a structured, maintainable and user-friendly solution for the Faculty of Technology.